Sequence and Progression: Questions and Full Solutions

Question 01

English

Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression and let $b_1, b_2, b_3, \ldots$ be a geometric progression with positive terms such that $a_1 = b_1 = 1$ and $a_3 = b_3$. Among all such pairs of progressions, the value of $a_2 + \dfrac{8}{b_2}$ is least for exactly one pair. For that pair, $a_{10} + b_4$ is equal to

(A) $\dfrac{41}{2}$
(B) 21
(C) $\dfrac{45}{2}$
(D) 24

Only one option is correct.

हिन्दी

माना $a_1, a_2, a_3, \ldots$ एक समान्तर श्रेढ़ी है तथा $b_1, b_2, b_3, \ldots$ धनात्मक पदों वाली एक गुणोत्तर श्रेढ़ी है, जिनके लिए $a_1 = b_1 = 1$ तथा $a_3 = b_3$ है। ऐसी सभी श्रेढ़ियों के युग्मों में से ठीक एक युग्म के लिए $a_2 + \dfrac{8}{b_2}$ का मान न्यूनतम है। उस युग्म के लिए $a_{10} + b_4$ बराबर है

(A) $\dfrac{41}{2}$
(B) 21
(C) $\dfrac{45}{2}$
(D) 24

केवल एक विकल्प सही है।

Solution

Let the arithmetic progression have common difference d and the geometric progression have common ratio r; since its terms are positive, $r > 0$.

$a_3 = 1 + 2d$ and $b_3 = r^2$, so $a_3 = b_3$ gives $1 + 2d = r^2$, that is, $d = \dfrac{r^2 - 1}{2}$.

Hence $a_2 = 1 + d = \dfrac{1 + r^2}{2}$, which is the arithmetic mean of $a_1 = 1$ and $a_3 = r^2$, and $b_2 = r$, which is the geometric mean of $b_1 = 1$ and $b_3 = r^2$.

The expression to be minimised is $E(r) = \dfrac{1 + r^2}{2} + \dfrac{8}{r}$, $r > 0$.

A two-term AM-GM on $\dfrac{r^2}{2}$ and $\dfrac{8}{r}$ does not give a constant product, so split $\dfrac{8}{r}$ into two equal parts: $E(r) = \dfrac{1}{2} + \dfrac{r^2}{2} + \dfrac{4}{r} + \dfrac{4}{r}$.

The product $\dfrac{r^2}{2} \cdot \dfrac{4}{r} \cdot \dfrac{4}{r} = 8$ is constant, so by AM-GM for three positive numbers, $\dfrac{r^2}{2} + \dfrac{4}{r} + \dfrac{4}{r} \geq 3\sqrt[3]{8} = 6$.

Therefore $E(r) \geq \dfrac{1}{2} + 6 = \dfrac{13}{2}$, with equality if and only if $\dfrac{r^2}{2} = \dfrac{4}{r}$, that is, $r^3 = 8$, $r = 2$. (Check by calculus: $E'(r) = r - \dfrac{8}{r^2}$ changes sign from negative to positive at $r = 2$.)

So exactly one pair of progressions gives the least value, the one with $r = 2$ and $d = \dfrac{4-1}{2} = \dfrac{3}{2}$.

For this pair $a_{10} = a_1 + 9d = 1 + \dfrac{27}{2} = \dfrac{29}{2}$ and $b_4 = r^3 = 8$.

$a_{10} + b_4 = \dfrac{29}{2} + 8 = \dfrac{45}{2}$. (Taking $a_{10} = a_1 + 10d$ by mistake gives 24.)

Correct Option: (C)

Question 02

English

Let a and b be positive real numbers whose geometric mean is 2, and let A and H denote the arithmetic mean and the harmonic mean of a and b, respectively. Over all such pairs, $A + 4H$ attains its minimum value. For a pair at which this minimum is attained, the ratio of the larger of a, b to the smaller is

(A) $7 + 4\sqrt{3}$
(B) $2 + \sqrt{3}$
(C) $3 + 2\sqrt{2}$
(D) 4

Only one option is correct.

हिन्दी

माना a तथा b ऐसी धनात्मक वास्तविक संख्याएँ हैं जिनका गुणोत्तर माध्य 2 है, तथा A और H क्रमशः a और b के समान्तर माध्य और हरात्मक माध्य हैं। ऐसे सभी युग्मों पर $A + 4H$ अपना न्यूनतम मान प्राप्त करता है। जिस युग्म के लिए यह न्यूनतम मान प्राप्त होता है, उसके लिए a, b में से बड़ी संख्या और छोटी संख्या का अनुपात है

(A) $7 + 4\sqrt{3}$
(B) $2 + \sqrt{3}$
(C) $3 + 2\sqrt{2}$
(D) 4

केवल एक विकल्प सही है।

Solution

For positive a, b: $A = \dfrac{a+b}{2}$, $G = \sqrt{ab}$, $H = \dfrac{2ab}{a+b}$. Multiplying, $AH = ab = G^2$.

Here $G = 2$, so $AH = 4$ for every admissible pair.

A and $4H$ are positive with constant product $4AH = 16$, so by AM-GM $A + 4H \geq 2\sqrt{16} = 8$.

Equality holds if and only if $A = 4H$. With $AH = 4$ this gives $4H^2 = 4$, so $H = 1$ and $A = 4$. These values satisfy $A \geq G \geq H$ $(4 \geq 2 \geq 1)$, so the minimum 8 is attained.

From $A = 4$: $a + b = 8$; from $G = 2$: $ab = 4$. So a and b are the roots of $t^2 - 8t + 4 = 0$, namely $t = 4 \pm 2\sqrt{3}$; both are positive and they are unequal.

The required ratio is $\dfrac{4 + 2\sqrt{3}}{4 - 2\sqrt{3}}$. Multiplying numerator and denominator by $4 + 2\sqrt{3}$: $\dfrac{(4 + 2\sqrt{3})^2}{16 - 12} = \dfrac{28 + 16\sqrt{3}}{4} = 7 + 4\sqrt{3}$.

(The value $2 + \sqrt{3}$ is its square root, and 4 is the ratio $A : H$, not $a : b$.)

Correct Option: (A)

Question 03

English

In a hall the rows of seats are numbered $1, 2, 3, \ldots$. The first row has 4 seats, every row has more seats than the row before it, and the increases in the number of seats from one row to the next form an arithmetic progression. The third row has 18 seats and the sixth row has 69 seats. Then the total number of seats in the first 10 rows is

(A) 715
(B) 745
(C) 760
(D) 790

Only one option is correct.

हिन्दी

एक सभागार में सीटों की पंक्तियाँ $1, 2, 3, \ldots$ क्रमांकित हैं। पहली पंक्ति में 4 सीटें हैं, प्रत्येक पंक्ति में उससे पहले वाली पंक्ति से अधिक सीटें हैं, तथा एक पंक्ति से अगली पंक्ति में सीटों की संख्या में होने वाली वृद्धियाँ एक समान्तर श्रेढ़ी बनाती हैं। तीसरी पंक्ति में 18 सीटें तथा छठी पंक्ति में 69 सीटें हैं। तब प्रथम 10 पंक्तियों में सीटों की कुल संख्या है

(A) 715
(B) 745
(C) 760
(D) 790

केवल एक विकल्प सही है।

Solution

Let u_n be the number of seats in row n and let the increases be $u_2 - u_1 = p$, $u_3 - u_2 = p + q$, and in general $u_{n+1} - u_n = p + (n-1)q$, an arithmetic progression with first term p and common difference q.

Adding the first $n-1$ increases: $u_n = u_1 + \displaystyle\sum_{k=1}^{n-1} \big(p + (k-1)q\big) = 4 + (n-1)p + \frac{(n-1)(n-2)}{2} q.$

Row 3: $u_3 = 4 + 2p + q = 18$, so $2p + q = 14$.

Row 6: $u_6 = 4 + 5p + 10q = 69$, so $5p + 10q = 65$, that is, $p + 2q = 13$.

From $q = 14 - 2p$: $p + 28 - 4p = 13 \Rightarrow p = 5$, and then $q = 4$. The increases $5, 9, 13, \ldots$ are positive, as required.

Hence $u_n = 4 + 5(n-1) + 2(n-1)(n-2) = 2n^2 - n + 3$; the first six rows have $4, 9, 18, 31, 48, 69$ seats, which matches the data.

Total in 10 rows: $\displaystyle\sum_{n=1}^{10} (2n^2 - n + 3) = 2 \cdot \frac{10 \cdot 11 \cdot 21}{6} - \frac{10 \cdot 11}{2} + 30 = 770 - 55 + 30 = 745.$

Correct Option: (B)

Question 04

English

Let $a_1, a_2, a_3, \ldots$ be an infinite geometric progression of non-zero real numbers with $a_1 = 2$ and common ratio r, where $0 < |r| < 1$. The line $a_1x + a_2y + a_3 = 0$ touches the circle $12\left(x^2 + y^2\right) = 1$. Then $a_1^2 + a_2^2 + a_3^2 + \cdots$ is equal to

(A) 3
(B) $3 + \sqrt{3}$
(C) $\dfrac{9}{2}$
(D) 6

Only one option is correct.

हिन्दी

माना $a_1, a_2, a_3, \ldots$ शून्येतर वास्तविक संख्याओं की एक अनन्त गुणोत्तर श्रेढ़ी है, जिसमें $a_1 = 2$ तथा सार्व अनुपात r है, जहाँ $0 < |r| < 1$ है। रेखा $a_1x + a_2y + a_3 = 0$ वृत्त $12\left(x^2 + y^2\right) = 1$ को स्पर्श करती है। तब $a_1^2 + a_2^2 + a_3^2 + \cdots$ बराबर है

(A) 3
(B) $3 + \sqrt{3}$
(C) $\dfrac{9}{2}$
(D) 6

केवल एक विकल्प सही है।

Solution

With $a_2 = 2r$ and $a_3 = 2r^2$, the line is $2x + 2ry + 2r^2 = 0$, that is, $x + ry + r^2 = 0$.

The circle $x^2 + y^2 = \dfrac{1}{12}$ has centre $(0, 0)$ and radius $\dfrac{1}{2\sqrt{3}}$.

The line touches the circle exactly when its distance from the centre equals the radius: $\dfrac{|r^2|}{\sqrt{1 + r^2}} = \dfrac{1}{2\sqrt{3}}$.

Squaring: $\dfrac{r^4}{1 + r^2} = \dfrac{1}{12} \Rightarrow 12r^4 - r^2 - 1 = 0 \Rightarrow (3r^2 - 1)(4r^2 + 1) = 0$.

Since $4r^2 + 1 > 0$, $r^2 = \dfrac{1}{3}$, so $r = \pm\dfrac{1}{\sqrt{3}}$; both satisfy $0 < |r| < 1$.

The squares $a_n^2 = 4r^{2(n-1)}$ form a geometric progression with first term 4 and common ratio $r^2 = \dfrac{1}{3}$, whatever the sign of r.

Its sum is $\dfrac{4}{1 - \frac{1}{3}} = 6$.

(The sum $a_1 + a_2 + \cdots = \dfrac{2}{1 - r}$ would be $3 + \sqrt{3}$ or $3 - \sqrt{3}$ depending on the sign, so it is not determined; the value 3 comes from using a_1 instead of a_1^2 as the first term.)

Correct Option: (D)

Question 05

English

Consider all vectors $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, with x, y, z real, satisfying $\vec{r} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 7$. Among these vectors let $\vec{r_0}$ be the one for which $x^2 + \frac{y^2}{4} + \frac{z^2}{9}$ attains its minimum value, and let θ be the angle between $\vec{r_0}$ and $\hat{i} + \hat{j} + \hat{k}$. Then $\tan^2\theta$ is equal to

(A) $\frac{1}{2}$
(B) 2
(C) $\frac{1}{3}$
(D) $\frac{3}{2}$

Only one option is correct.

हिन्दी

उन सभी सदिशों $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ पर विचार कीजिए, जहाँ x, y, z वास्तविक हैं तथा $\vec{r} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 7$ है। इन सदिशों में से माना $\vec{r_0}$ वह सदिश है जिसके लिए $x^2 + \frac{y^2}{4} + \frac{z^2}{9}$ अपना न्यूनतम मान प्राप्त करता है, तथा θ सदिशों $\vec{r_0}$ और $\hat{i} + \hat{j} + \hat{k}$ के बीच का कोण है। तब $\tan^2\theta$ बराबर है

(A) $\frac{1}{2}$
(B) 2
(C) $\frac{1}{3}$
(D) $\frac{3}{2}$

केवल एक विकल्प सही है।

Solution

Write the constraint as a dot product with weights that match the objective: let $\vec{u} = x\hat{i} + \frac{y}{2}\hat{j} + \frac{z}{3}\hat{k}$ and $\vec{v} = \hat{i} + 2\hat{j} + 3\hat{k}$. Then $\vec{u} \cdot \vec{v} = x + y + z = \vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 7$ and $|\vec{u}|^2 = x^2 + \frac{y^2}{4} + \frac{z^2}{9}$.

By the Cauchy-Schwarz inequality, $(\vec{u} \cdot \vec{v})^2 \le |\vec{u}|^2 |\vec{v}|^2$, so $49 \le 14\left(x^2 + \frac{y^2}{4} + \frac{z^2}{9}\right)$, that is, the expression is at least $\frac{7}{2}$.

Equality holds exactly when $\vec{u}$ is parallel to $\vec{v}$: $x = \lambda$, $\frac{y}{2} = 2\lambda$, $\frac{z}{3} = 3\lambda$, so $(x, y, z) = (\lambda, 4\lambda, 9\lambda)$.

The constraint gives $\lambda + 4\lambda + 9\lambda = 7$, so $\lambda = \frac{1}{2}$ and $\vec{r_0} = \frac{1}{2}\hat{i} + 2\hat{j} + \frac{9}{2}\hat{k}$ is the unique minimiser.

$|\vec{r_0}|^2 = \frac{1}{4} + 4 + \frac{81}{4} = \frac{49}{2}$, so $|\vec{r_0}| = \frac{7}{\sqrt{2}}$, and $|\hat{i} + \hat{j} + \hat{k}| = \sqrt{3}$.

$\cos\theta = \frac{\vec{r_0} \cdot (\hat{i} + \hat{j} + \hat{k})}{|\vec{r_0}|\sqrt{3}} = \frac{7}{\frac{7}{\sqrt{2}}\sqrt{3}} = \sqrt{\frac{2}{3}}$.

$\tan^2\theta = \sec^2\theta - 1 = \frac{3}{2} - 1 = \frac{1}{2}$. (The values 2 and $\frac{3}{2}$ are $\cot^2\theta$ and $\sec^2\theta$.)

Correct Option: (A)

Question 06

English

Let $a_1, a_2, \ldots, a_{40}$ be the arithmetic progression with $a_1 = 20$ and common difference -3, and let $b_1, b_2, \ldots, b_{40}$ be the arithmetic progression with $b_1 = -50$ and common difference 4. Then the sum of those terms of the first progression that are not terms of the second progression is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $a_1, a_2, \ldots, a_{40}$ वह समान्तर श्रेढ़ी है जिसमें $a_1 = 20$ तथा सार्व अन्तर -3 है, तथा $b_1, b_2, \ldots, b_{40}$ वह समान्तर श्रेढ़ी है जिसमें $b_1 = -50$ तथा सार्व अन्तर 4 है। तब प्रथम श्रेढ़ी के उन पदों का योग, जो द्वितीय श्रेढ़ी के पद नहीं हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The first progression is $20, 17, 14, \ldots$ with $a_{40} = 20 + 39(-3) = -97$; its sum is $\frac{40}{2}(20 + (-97)) = -1540$.

The second progression is $-50, -46, -42, \ldots$ with $b_{40} = -50 + 39 \cdot 4 = 106$.

Every $a_k = 20 - 3(k-1)$ leaves remainder 2 on division by 3, and every $b_k = -50 + 4(k-1)$ leaves remainder 2 on division by 4. A number common to both is therefore $\equiv 2 \pmod 3$ and $\equiv 2 \pmod 4$, that is, $\equiv 2 \pmod{12}$.

Conversely, a number $\equiv 2 \pmod{12}$ is a term of the first progression if it lies in $[-97, 20]$ and a term of the second if it lies in $[-50, 106]$. So the common terms are the numbers $\equiv 2 \pmod{12}$ in $[-50, 20]$.

These are $-46, -34, -22, -10, 2, 14$: an arithmetic progression of 6 terms with common difference 12 (the l.c.m. of 3 and 4). Their sum is $\frac{6}{2}(-46 + 14) = -96$.

The terms of the first progression that are not terms of the second therefore sum to $-1540 - (-96) = -1444$.

Final Answer: -1444

Question 07

English

Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression of real numbers such that $a_1a_2 = 10$, $a_2a_3 = 40$ and $a_3a_4 = 88$. Then the value of $|a_{12}|$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $a_1, a_2, a_3, \ldots$ वास्तविक संख्याओं की एक समान्तर श्रेढ़ी है, जिसके लिए $a_1a_2 = 10$, $a_2a_3 = 40$ तथा $a_3a_4 = 88$ है। तब $|a_{12}|$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let d be the common difference. For any n, $a_{n+1}a_{n+2} - a_na_{n+1} = a_{n+1}(a_{n+2} - a_n) = a_{n+1} \cdot 2d$.

So the products of consecutive terms have differences that again form an arithmetic progression: $2d\,a_2,\ 2d\,a_3, \ldots$ with common difference $2d \cdot d = 2d^2$.

From the data: $a_2a_3 - a_1a_2 = 30 = 2d\,a_2$ and $a_3a_4 - a_2a_3 = 48 = 2d\,a_3$.

Subtracting: $48 - 30 = 2d(a_3 - a_2) = 2d^2$, so $d^2 = 9$ and $d = \pm 3$.

If $d = 3$: $a_2 = \dfrac{30}{6} = 5$, $a_1 = 2$, $a_3 = 8$, $a_4 = 11$, and $2 \cdot 5 = 10$, $5 \cdot 8 = 40$, $8 \cdot 11 = 88$ hold.

If $d = -3$: $a_2 = -5$, $a_1 = -2$, $a_3 = -8$, $a_4 = -11$; the products are the same.

In both cases $a_{12} = a_1 + 11d = \pm(2 + 33) = \pm 35$, so $|a_{12}| = 35$.

Final Answer: 35

Question 08

English

Let $a_1, a_2, \ldots, a_9$ be nine consecutive terms of a decreasing arithmetic progression. The mean of these nine numbers is positive, their variance is 15, and $a_1a_9 = -\dfrac{95}{4}$. Then the value of $a_8 + a_9$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $a_1, a_2, \ldots, a_9$ एक ह्रासमान समान्तर श्रेढ़ी के नौ क्रमागत पद हैं। इन नौ संख्याओं का माध्य धनात्मक है, इनका प्रसरण 15 है, तथा $a_1a_9 = -\dfrac{95}{4}$ है। तब $a_8 + a_9$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For nine terms of an arithmetic progression the mean is the middle term: $m = a_5$. Write $a_k = m + (k-5)d$, where $d < 0$ because the progression is decreasing.

The deviations from the mean are $(k-5)d$, $k = 1, \ldots, 9$, so the variance is $\dfrac{d^2}{9}\displaystyle\sum_{j=-4}^{4} j^2 = \dfrac{d^2}{9} \cdot 60 = \dfrac{20d^2}{3}$

(the formula $\dfrac{(n^2-1)d^2}{12}$ with $n = 9$).

$\dfrac{20d^2}{3} = 15 \Rightarrow d^2 = \dfrac{9}{4} \Rightarrow d = -\dfrac{3}{2}$.

$a_1 = m - 4d$ and $a_9 = m + 4d$, so $a_1a_9 = m^2 - 16d^2 = m^2 - 36$.

$m^2 - 36 = -\frac{95}{4} \Rightarrow m^2 = \frac{49}{4}$; the mean is positive, so $m = \frac{7}{2}$.

Then $a_8 = m + 3d = \frac{7}{2} - \frac{9}{2} = -1$ and $a_9 = m + 4d = \frac{7}{2} - 6 = -\frac{5}{2}$.

$a_8 + a_9 = -1 - \frac{5}{2} = -3.5$.

Final Answer: -3.5

Question 09

English

Let $a_1, a_2, a_3, \ldots$ be a decreasing arithmetic progression such that $\sum_{n=1}^{\infty} \frac{a_n}{(n-1)!} = \frac{3e}{2}$ and $\sum_{n=1}^{\infty} \frac{a_n^2}{(n-1)!} = \frac{37e}{16}$. Then the value of a_7 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $a_1, a_2, a_3, \ldots$ एक ह्रासमान समान्तर श्रेढ़ी है जिसके लिए $\sum_{n=1}^{\infty} \frac{a_n}{(n-1)!} = \frac{3e}{2}$ तथा $\sum_{n=1}^{\infty} \frac{a_n^2}{(n-1)!} = \frac{37e}{16}$ है। तब a_7 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $a = a_1$ and let d be the common difference ($d < 0$). With $m = n - 1$, $a_n = a + md$ and the sums run over $m = 0, 1, 2, \ldots$

From $e^x = \sum_{m \geq 0} \frac{x^m}{m!}$ at $x = 1$: $\sum \frac{1}{m!} = e$; $\sum \frac{m}{m!} = \sum_{m \geq 1} \frac{1}{(m-1)!} = e$; $\sum \frac{m^2}{m!} = \sum \frac{m(m-1)}{m!} + \sum \frac{m}{m!} = \sum_{m \geq 2} \frac{1}{(m-2)!} + e = 2e$.

First condition: $\sum \frac{a + md}{m!} = ae + de = e(a + d) = \frac{3e}{2}$, so $a + d = \frac{3}{2}$.

Second condition: $\sum \frac{a^2 + 2amd + m^2 d^2}{m!} = a^2 e + 2ade + 2d^2 e = e\big((a+d)^2 + d^2\big)$.

So $(a + d)^2 + d^2 = \frac{37}{16}$, i.e. $\frac{36}{16} + d^2 = \frac{37}{16}$ and $d^2 = \frac{1}{16}$.

The progression is decreasing, so $d = -\frac{1}{4}$ and $a = \frac{3}{2} + \frac{1}{4} = \frac{7}{4}$.

$a_7 = a + 6d = \frac{7}{4} - \frac{3}{2} = \frac{1}{4} = 0.25$.

Final Answer: 0.25

Question 10

English

Let a and b be distinct positive real numbers. Numbers A_1, A_2, A_3, positive numbers G_1, G_2, G_3, and numbers H_1, H_2, H_3 are inserted between a and b such that a, A_1, A_2, A_3, b are in arithmetic progression, a, G_1, G_2, G_3, b are in geometric progression, and a, H_1, H_2, H_3, b are in harmonic progression. Suppose that $G_1G_2G_3 = 1000$. Then the minimum value of $A_1 + 36H_3$, taken over all such pairs (a, b), is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना a तथा b भिन्न धनात्मक वास्तविक संख्याएँ हैं। a और b के बीच संख्याएँ A_1, A_2, A_3, धनात्मक संख्याएँ G_1, G_2, G_3 तथा संख्याएँ H_1, H_2, H_3 इस प्रकार रखी जाती हैं कि a, A_1, A_2, A_3, b समान्तर श्रेढ़ी में, a, G_1, G_2, G_3, b गुणोत्तर श्रेढ़ी में तथा a, H_1, H_2, H_3, b हरात्मक श्रेढ़ी में हैं। माना $G_1G_2G_3 = 1000$ है। तब ऐसे सभी युग्मों (a, b) पर $A_1 + 36H_3$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

In the geometric progression a, G_1, G_2, G_3, b, terms equidistant from the ends have equal products: $G_1G_3 = G_2^2 = ab$. So $G_1G_2G_3 = G_2^3 = 1000$, $G_2 = 10$ and $ab = 100$.

The arithmetic progression has common difference $\dfrac{b-a}{4}$, so $A_1 = a + \dfrac{b-a}{4} = \dfrac{3a+b}{4}$.

In the harmonic progression the reciprocals $\dfrac{1}{a}, \dfrac{1}{H_1}, \dfrac{1}{H_2}, \dfrac{1}{H_3}, \dfrac{1}{b}$ are in arithmetic progression, so $\dfrac{1}{H_3} = \dfrac{1}{a} + \dfrac{3}{4}\left(\dfrac{1}{b} - \dfrac{1}{a}\right) = \dfrac{b+3a}{4ab}$ and $H_3 = \dfrac{4ab}{3a+b}$.

Hence $A_1H_3 = ab = 100$ (the AM-HM pairing $A_kH_{4-k} = ab$).

$A_1 + 36H_3 = A_1 + \dfrac{3600}{A_1}$. By AM-GM, $A_1 + \dfrac{3600}{A_1} \geq 2\sqrt{3600} = 120$, with equality iff $A_1 = 60$.

Attainment: $A_1 = 60$ means $3a+b = 240$; with $ab = 100$, $a(240-3a) = 100$, i.e. $3a^2-240a+100 = 0$. Its discriminant $240^2 - 1200 > 0$, both roots are positive and less than 80, so $b = 240 - 3a > 0$; and $a = b$ would force $a = 60$, $ab = 3600 \neq 100$. So a valid pair exists.

The minimum value of $A_1 + 36H_3$ is 120.

Final Answer: 120

Question 11

English

Let $a_1, a_2, \ldots, a_{100}$ be an increasing arithmetic progression such that $a_4 + a_{12} = 50$ and $a_3 a_{13} = 400$, and let $b_1, b_2, \ldots, b_{100}$ be an arithmetic progression such that $b_2 + b_{18} = 100$ and $b_5 + b_6 + b_7 = 90$. The numbers that are terms of both progressions are written as $c_1 < c_2 < \cdots < c_m$. Then the value of $\dfrac{c_1 + c_2 + \cdots + c_m}{m}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $a_1, a_2, \ldots, a_{100}$ एक वर्धमान समान्तर श्रेढ़ी है जिसके लिए $a_4 + a_{12} = 50$ तथा $a_3 a_{13} = 400$ है, तथा $b_1, b_2, \ldots, b_{100}$ एक समान्तर श्रेढ़ी है जिसके लिए $b_2 + b_{18} = 100$ तथा $b_5 + b_6 + b_7 = 90$ है। जो संख्याएँ दोनों श्रेढ़ियों के पद हैं, उन्हें $c_1 < c_2 < \cdots < c_m$ के रूप में लिखा जाता है। तब $\dfrac{c_1 + c_2 + \cdots + c_m}{m}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

In an arithmetic progression $a_p + a_q = 2a_r$ whenever $p + q = 2r$. So $a_4 + a_{12} = 2a_8 = 50$ and $a_8 = 25$.

With common difference d: $a_3 = 25 - 5d$ and $a_{13} = 25 + 5d$, so $a_3 a_{13} = 625 - 25d^2 = 400$, $d^2 = 9$; the progression is increasing, so $d = 3$.

Then $a_n = a_8 + 3(n - 8) = 3n + 1$, so the first progression is $4, 7, 10, \ldots, 301$.

Similarly $b_2 + b_{18} = 2b_{10} = 100$ gives $b_{10} = 50$, and $b_5 + b_6 + b_7 = 3b_6 = 90$ gives $b_6 = 30$. The common difference is $\dfrac{50 - 30}{4} = 5$, so $b_n = 5n$: $5, 10, \ldots, 500$.

A common term c satisfies $c \equiv 1 \pmod 3$ and $c \equiv 0 \pmod 5$, that is, $c \equiv 10 \pmod{15}$; it must also lie in $[4, 301] \cap [5, 500] = [5, 301]$.

Every number $\equiv 10 \pmod{15}$ in this range is a term of both progressions, so the common terms are $10, 25, 40, \ldots, 295$, an arithmetic progression with common difference 15 and $m = \dfrac{295 - 10}{15} + 1 = 20$ terms.

For an arithmetic progression, sum $= \dfrac{m}{2}(c_1 + c_m)$, so the mean is $\dfrac{c_1 + c_m}{2} = \dfrac{10 + 295}{2} = 152.5$.

Final Answer: 152.5

Question 12

English

For positive integers p and q, consider the arithmetic progressions $7,\ 7+p,\ 7+2p,\ \ldots$ and $7,\ 7+q,\ 7+2q,\ \ldots$. Let S be the set of all ordered pairs (p,q) for which the numbers common to the two progressions form an arithmetic progression with common difference 36. Then the arithmetic mean of the numbers $p+q$, taken over all pairs $(p,q) \in S$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

धनात्मक पूर्णांकों p तथा q के लिए समान्तर श्रेढ़ियों $7,\ 7+p,\ 7+2p,\ \ldots$ तथा $7,\ 7+q,\ 7+2q,\ \ldots$ पर विचार कीजिए। माना S उन सभी क्रमित युग्मों (p,q) का समुच्चय है जिनके लिए दोनों श्रेढ़ियों में उभयनिष्ठ संख्याएँ 36 सार्व अन्तर वाली एक समान्तर श्रेढ़ी बनाती हैं। तब सभी युग्मों $(p,q) \in S$ पर ली गई संख्याओं $p+q$ का समान्तर माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A number $7+N$ ($N \geq 0$) is a term of the first progression iff $p \mid N$, and of the second iff $q \mid N$. So it is common iff $\text{lcm}(p,q) \mid N$, and the common terms are $7,\ 7+L,\ 7+2L, \ldots$ with $L = \text{lcm}(p,q)$.

Hence $(p,q) \in S$ exactly when $\text{lcm}(p,q) = 36 = 2^2 \cdot 3^2$; in particular $p \mid 36$ and $q \mid 36$.

Write $p = 2^i 3^j$ and $q = 2^k 3^l$ with exponents in $\{0,1,2\}$. The condition is $\max(i,k) = 2$ and $\max(j,l) = 2$.

For the prime 2 the pairs (i,k) with maximum 2 are $(2,0),(2,1),(2,2),(0,2),(1,2)$: five; the same for the prime 3. So $|S| = 25$.

Fix $p = 2^i 3^j$. If $i = 2$, k can be $0,1,2$ (3 choices); if $i < 2$, $k = 2$ (1 choice). So the number of q is $c_i c_j$ with $c_0 = c_1 = 1$, $c_2 = 3$.

$$\sum_{(p,q)\in S} p = \sum_{i,j} 2^i 3^j c_i c_j = (1+2+3\cdot 4)\,(1+3+3\cdot 9) = 15 \cdot 31 = 465.$$

The set S is symmetric in p and q, so $\sum_S q = 465$ too, and the mean of $p+q$ is $\dfrac{465+465}{25} = 37.2$.

Final Answer: 37.2

Question 13

English

Let a be a real number with $|a| < 2$ such that $\sum_{n=0}^{\infty} \frac{1 + a + a^2 + \cdots + a^n}{2^n} = \frac{8}{5}$. Then the value of a is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना a एक वास्तविक संख्या है, जहाँ $|a| < 2$ है, तथा $\sum_{n=0}^{\infty} \frac{1 + a + a^2 + \cdots + a^n}{2^n} = \frac{8}{5}$ है। तब a का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If $a = 1$, the n-th numerator is $n + 1$ and the sum is $\sum_{n\geq 0} \frac{n+1}{2^n} = 4 \neq \frac{8}{5}$, so $a \neq 1$.

For $a \neq 1$ the numerator is a finite GP: $1 + a + \cdots + a^n = \frac{1 - a^{n+1}}{1 - a}$.

So the given series is $\frac{1}{1-a}\left[\sum_{n\geq 0} \frac{1}{2^n} - a\sum_{n\geq 0}\left(\frac{a}{2}\right)^n\right]$; both are convergent GPs because $\left|\frac{a}{2}\right| < 1$.

$\sum_{n\geq 0} \frac{1}{2^n} = 2$ and $\sum_{n\geq 0}\left(\frac{a}{2}\right)^n = \frac{2}{2-a}$.

Hence the sum is $\frac{1}{1-a}\left[2 - \frac{2a}{2-a}\right] = \frac{1}{1-a} \cdot \frac{4-4a}{2-a} = \frac{4}{2-a}$.

$\frac{4}{2-a} = \frac{8}{5}$ gives $2 - a = \frac{5}{2}$, so $a = -\frac{1}{2}$, which satisfies $|a| < 2$.

Final Answer: -0.5

Question 14

English

Let $b_1, b_2, b_3, \ldots$ be a geometric progression of real numbers such that $\sum_{n=1}^{\infty} b_n = 6$ and $\sum_{n=1}^{\infty} nb_n = 12$.

Then the value of $\sum_{n=1}^{\infty} n^2 b_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $b_1, b_2, b_3, \ldots$ वास्तविक संख्याओं की एक गुणोत्तर श्रेढ़ी है जिसके लिए $\sum_{n=1}^{\infty} b_n = 6$ तथा $\sum_{n=1}^{\infty} nb_n = 12$ है। तब $\sum_{n=1}^{\infty} n^2 b_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $b_n = br^{n-1}$. Convergence of $\sum b_n$ forces $|r| < 1$, and then $\sum b_n = \dfrac{b}{1-r} = 6$.

$\sum nb_n = b\sum nr^{n-1}$ is an arithmetico-geometric series: with $U = \sum nr^{n-1}$, $U - rU = \sum r^{n-1} = \dfrac{1}{1-r}$, so $U = \dfrac{1}{(1-r)^2}$ and $\sum nb_n = \dfrac{b}{(1-r)^2} = 12$.

Dividing, $\dfrac{1}{1-r} = 2$, so $r = \dfrac{1}{2}$ and $b = 6(1-r) = 3$.

For $T = \sum n^2 r^{n-1}$: $T - rT = \sum \left(n^2 - (n-1)^2\right) r^{n-1} = \sum (2n-1) r^{n-1} = 2U - \dfrac{1}{1-r} = \dfrac{2}{(1-r)^2} - \dfrac{1}{1-r} = \dfrac{1+r}{(1-r)^2}$.

So $T = \dfrac{1+r}{(1-r)^3} = \dfrac{3/2}{1/8} = 12$.

$\sum n^2 b_n = bT = 3 \cdot 12 = 36$.

Final Answer: 36

Question 15

English

Let $z = \cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}$, where $i = \sqrt{-1}$. Then the value of $\displaystyle\sum_{k=0}^{11}\left|1+z^k\right|^8$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $z = \cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}$ है, जहाँ $i = \sqrt{-1}$ है। तब $\displaystyle\sum_{k=0}^{11}\left|1+z^k\right|^8$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By De Moivre's theorem $z^k = \cos\dfrac{k\pi}{6} + i\sin\dfrac{k\pi}{6}$, so $z^{12} = 1$ and $z^m = 1$ exactly when $12 \mid m$. Also $|z| = 1$, so $\bar{z} = z^{-1}$.

For any complex w, $|w|^2 = w\bar{w}$. Hence $\left|1+z^k\right|^8 = \left((1+z^k)(1+z^{-k})\right)^4 = (1+z^k)^4 \cdot \dfrac{(z^k+1)^4}{z^{4k}} = z^{-4k}(1+z^k)^8$.

Expanding by the binomial theorem: $\displaystyle\sum_{k=0}^{11} z^{-4k}\sum_{j=0}^{8}\binom{8}{j}z^{kj} = \sum_{j=0}^{8}\binom{8}{j}\sum_{k=0}^{11}\left(z^{j-4}\right)^k$.

The inner sum is a geometric progression of 12 terms with ratio $w = z^{j-4}$. If $w = 1$ it equals 12; otherwise it equals $\dfrac{1-w^{12}}{1-w} = 0$ because $w^{12} = \left(z^{12}\right)^{j-4} = 1$.

$w = 1$ needs $12 \mid (j-4)$; as $j - 4 \in \{-4, \ldots, 4\}$ this happens only for $j = 4$.

So the sum is $12\binom{8}{4} = 12 \cdot 70 = 840$. (Direct check: $\left|1+z^k\right| = 2\left|\cos\dfrac{k\pi}{12}\right|$ and $\displaystyle\sum_{k=0}^{11} 256\cos^8\frac{k\pi}{12} = 840$.)

Final Answer: 840

Question 16

English

Two players A and B throw a fair die alternately, A throwing first. A wins the game the moment A throws a 6, and B wins the game the moment B throws a 5 or a 6; in addition, whichever player throws a 1 loses the game at once, the other player being the winner. The throws continue until the game is decided, and the loser then pays the winner Rs. 60. Then the expected gain, in rupees, of A is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दो खिलाड़ी A तथा B बारी-बारी से एक अनभिनत पासा फेंकते हैं, पहले A फेंकता है। जैसे ही A के पासे पर 6 आता है, A खेल जीत जाता है, तथा जैसे ही B के पासे पर 5 या 6 आता है, B खेल जीत जाता है; इसके अतिरिक्त, जिस भी खिलाड़ी के पासे पर 1 आता है वह तुरन्त खेल हार जाता है और दूसरा खिलाड़ी विजेता होता है। खेल का निर्णय होने तक फेंकना जारी रहता है, तथा फिर हारने वाला विजेता को $\text{Rs. } 60$ देता है। तब A का अपेक्षित लाभ (रुपयों में) _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

On a throw by A: a 6 (probability $\frac{1}{6}$) ends the game in A's favour, a 1 (probability $\frac{1}{6}$) ends it in B's favour, and $2, 3, 4, 5$ (probability $\frac{4}{6} = \frac{2}{3}$) pass the turn to B.

On a throw by B: 5 or 6 (probability $\frac{1}{3}$) ends it in B's favour, a 1 (probability $\frac{1}{6}$) ends it in A's favour, and $2, 3, 4$ (probability $\frac{1}{2}$) pass the turn back to A.

Call one throw by A followed (if needed) by one throw by B a round. A wins during a round with probability $\dfrac{1}{6} + \dfrac{2}{3} \cdot \dfrac{1}{6} = \dfrac{5}{18}$, and the round leaves the game undecided with probability $\dfrac{2}{3} \cdot \dfrac{1}{2} = \dfrac{1}{3}$.

Rounds are independent, so A wins in round $k+1$ with probability $\left(\dfrac{1}{3}\right)^k \dfrac{5}{18}$. Summing the geometric series: $P(A) = \dfrac{5}{18} \cdot \dfrac{1}{1 - \frac{1}{3}} = \dfrac{5}{12}$.

Similarly B wins during a round with probability $\dfrac{1}{6} + \dfrac{2}{3} \cdot \dfrac{1}{3} = \dfrac{7}{18}$, so $P(B) = \dfrac{7}{18} \cdot \dfrac{3}{2} = \dfrac{7}{12}$. Since $\dfrac{5}{12} + \dfrac{7}{12} = 1$, the game ends with probability 1.

A gains Rs. 60 with probability $\dfrac{5}{12}$ and loses Rs. 60 with probability $\dfrac{7}{12}$: expected gain $= 60 \cdot \dfrac{5}{12} - 60 \cdot \dfrac{7}{12} = -10$.

Final Answer: -10

Question 17

English

The value of $\sum_{n=1}^{\infty} \frac{n\left(1+2+2^2+\cdots+2^{n-1}\right)}{6^n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\sum_{n=1}^{\infty} \frac{n\left(1+2+2^2+\cdots+2^{n-1}\right)}{6^n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The bracket is a geometric progression with ratio 2: $1+2+\cdots+2^{n-1}=2^n-1$.

So the n-th term is $\frac{n(2^n-1)}{6^n} = n\left(\frac{1}{3}\right)^n - n\left(\frac{1}{6}\right)^n$: the sequence of terms is converted into the difference of two arithmetico-geometric sequences.

For $|x|<1$ let $S(x)=\sum_{n\geq 1} nx^n$. Then $S-xS=\sum_{n\geq 1} x^n = \frac{x}{1-x}$, so $S(x)=\frac{x}{(1-x)^2}$; both series converge, so they may be subtracted term by term.

$S\left(\frac{1}{3}\right) = \frac{1/3}{(2/3)^2} = \frac{3}{4}$ and $S\left(\frac{1}{6}\right) = \frac{1/6}{(5/6)^2} = \frac{6}{25}$.

The required sum is $\frac{3}{4} - \frac{6}{25} = \frac{75-24}{100} = \frac{51}{100} = 0.51$.

Final Answer: 0.51

Question 18

English

Let $\omega = \dfrac{-1+i\sqrt{3}}{2}$, where $i = \sqrt{-1}$, and let $\displaystyle\sum_{k=1}^{6}(1-\omega)^k = a + ib$, where a and b are real numbers. Then the value of $a + \sqrt{3}\,b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\omega = \dfrac{-1+i\sqrt{3}}{2}$ है, जहाँ $i = \sqrt{-1}$ है, तथा $\displaystyle\sum_{k=1}^{6}(1-\omega)^k = a + ib$ है, जहाँ a तथा b वास्तविक संख्याएँ हैं। तब $a + \sqrt{3}\,b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The terms $(1-\omega)^k$ form a geometric progression with ratio $z = 1-\omega \neq 1$, so $\displaystyle\sum_{k=1}^{6} z^k = \frac{z(z^6-1)}{z-1}$.

Here $z - 1 = -\omega$.

$z = 1-\omega = \dfrac{3}{2} - \dfrac{\sqrt{3}}{2}i$ has modulus $\sqrt{\dfrac{9}{4}+\dfrac{3}{4}} = \sqrt{3}$ and argument $-\dfrac{\pi}{6}$: $z = \sqrt{3}\left(\cos\dfrac{\pi}{6} - i\sin\dfrac{\pi}{6}\right)$.

By De Moivre's theorem $z^6 = (\sqrt{3})^6(\cos\pi - i\sin\pi) = -27$, so $z^6 - 1 = -28$.

Hence the sum is $\dfrac{-28z}{-\omega} = \dfrac{28(1-\omega)}{\omega}$. Since $\omega^3 = 1$, $\dfrac{1}{\omega} = \omega^2$, so the sum is $28(1-\omega)\omega^2 = 28(\omega^2 - \omega^3) = 28(\omega^2 - 1)$.

$\omega^2 = \dfrac{-1-i\sqrt{3}}{2}$, so $\omega^2 - 1 = -\dfrac{3}{2} - \dfrac{\sqrt{3}}{2}i$ and the sum is $-42 - 14\sqrt{3}\,i$.

So $a = -42$, $b = -14\sqrt{3}$, and $a + \sqrt{3}\,b = -42 - 42 = -84$.

Final Answer: -84

Question 19

English

For each real number $t > 0$, let $m(t)$ denote the minimum value of $x^2+y^2+z^2$ over all real numbers x, y, z satisfying $tx + y + \frac{4}{t}z = -\frac{9}{2}$. Let t_0 be the value of t for which $m(t)$ is greatest, and let (x_0, y_0, z_0) be the point at which $x^2+y^2+z^2$ attains the value $m(t_0)$ subject to $t_0x+y+\frac{4}{t_0}z = -\frac{9}{2}$. Then the value of $x_0 + y_0 + z_0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक वास्तविक संख्या $t > 0$ के लिए माना $m(t)$, $tx + y + \frac{4}{t}z = -\frac{9}{2}$ को संतुष्ट करने वाली सभी वास्तविक संख्याओं x, y, z पर $x^2+y^2+z^2$ का न्यूनतम मान है। माना t_0, t का वह मान है जिसके लिए $m(t)$ अधिकतम है, तथा (x_0, y_0, z_0) वह बिन्दु है जिस पर प्रतिबन्ध $t_0x + y + \frac{4}{t_0}z = -\frac{9}{2}$ के अन्तर्गत $x^2 + y^2 + z^2$ का मान $m(t_0)$ प्राप्त होता है। तब $x_0 + y_0 + z_0$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Fix $t > 0$ and put $\overrightarrow{p} = \left(t, 1, \frac{4}{t}\right)$. The condition says $\overrightarrow{p} \cdot (x, y, z) = -\frac{9}{2}$.

By the Cauchy-Schwarz inequality, $\frac{81}{4} \le |\overrightarrow{p}|^2(x^2 + y^2 + z^2)$, with equality iff (x, y, z) is a multiple of $\overrightarrow{p}$. So $m(t) = \frac{81}{4\left(t^2 + 1 + \frac{16}{t^2}\right)}$, attained only at $(x, y, z) = \lambda\overrightarrow{p}$ with $\lambda = -\frac{9}{2|\overrightarrow{p}|^2}$.

$m(t)$ is greatest when $t^2 + \frac{16}{t^2}$ is least. By AM-GM, $t^2 + \frac{16}{t^2} \ge 2\sqrt{16} = 8$, with equality iff $t^2 = 4$, i.e. $t = 2$ (as $t > 0$). So $t_0 = 2$ and $m(t_0) = \frac{81}{4 \cdot 9} = \frac{9}{4}$.

For $t_0 = 2$, $\overrightarrow{p} = (2, 1, 2)$, $|\overrightarrow{p}|^2 = 9$, so $\lambda = -\frac{9}{18} = -\frac{1}{2}$ and $(x_0, y_0, z_0) = \left(-1, -\frac{1}{2}, -1\right)$.

Check: $2(-1) + \left(-\frac{1}{2}\right) + 2(-1) = -\frac{9}{2}$ and $x_0^2 + y_0^2 + z_0^2 = \frac{9}{4}$.

$x_0 + y_0 + z_0 = -1 - \frac{1}{2} - 1 = -2.5$.

Final Answer: -2.5

Question 20

English

Let α and β be real numbers in $\left(0, \frac{\pi}{2}\right)$ such that $\alpha + \beta = \frac{\pi}{2}$. Then the minimum value of $\left(1 + 24\,\mathrm{cosec}^2\alpha\right)\left(1 + 2\,\mathrm{cosec}^2\beta\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना α तथा β, $\left(0, \frac{\pi}{2}\right)$ में ऐसी वास्तविक संख्याएँ हैं कि $\alpha + \beta = \frac{\pi}{2}$ है। तब $\left(1 + 24\,\mathrm{cosec}^2\alpha\right)\left(1 + 2\,\mathrm{cosec}^2\beta\right)$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

From $\alpha + \beta = \frac{\pi}{2}$: $\mathrm{cosec}\,\beta = \frac{1}{\sin\left(\frac{\pi}{2} - \alpha\right)} = \sec\alpha$, so the expression is $\left(1 + 24\,\mathrm{cosec}^2\alpha\right)\left(1 + 2\sec^2\alpha\right)$ with $\alpha \in \left(0, \frac{\pi}{2}\right)$.

Let $t = \tan^2\alpha$, which takes every value in $(0, \infty)$. Then $\mathrm{cosec}^2\alpha = 1 + \cot^2\alpha = 1 + \frac{1}{t}$ and $\sec^2\alpha = 1 + t$.

The expression becomes $\left(1 + 24 + \frac{24}{t}\right)(1 + 2 + 2t) = \left(25 + \frac{24}{t}\right)(3 + 2t) = 75 + 50t + \frac{72}{t} + 48 = 123 + 50t + \frac{72}{t}$.

$50t$ and $\frac{72}{t}$ are positive with constant product 3600, so by AM-GM $50t + \frac{72}{t} \geq 2\sqrt{3600} = 120$.

Equality holds iff $50t = \frac{72}{t}$, i.e. $t^2 = \frac{36}{25}$, $t = \frac{6}{5}$; this corresponds to $\tan\alpha = \sqrt{\frac{6}{5}}$, a valid α, with $\beta = \frac{\pi}{2} - \alpha$.

The minimum value is $123 + 120 = 243$. (Guessing the symmetric point $\alpha = \beta = \frac{\pi}{4}$ gives $49 \cdot 5 = 245$, which is not the minimum.)

Final Answer: 243

Question 21

English

Let A be a point on the positive x-axis and B a point on the positive y-axis such that the line segment AB touches the circle $x^2 + y^2 = 1$. Let O be the origin and let r be the radius of the circle inscribed in the triangle OAB. Then the minimum value of r, taken over all such positions of A and B, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना A धनात्मक x-अक्ष पर तथा B धनात्मक y-अक्ष पर ऐसे बिन्दु हैं कि रेखाखण्ड AB वृत्त $x^2 + y^2 = 1$ को स्पर्श करता है। माना O मूलबिन्दु है तथा r त्रिभुज OAB के अन्तर्वृत्त की त्रिज्या है। तब A और B की ऐसी सभी स्थितियों पर r का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $OA = a$, $OB = b$ and $AB = c$; the triangle OAB is right-angled at O, so $c^2 = a^2 + b^2$.

AB touches the unit circle centred at O, so the perpendicular distance from O to AB is 1. Twice the area of the triangle is $ab = c \cdot 1$, hence $c = ab$.

By AM-GM, $c^2 = a^2 + b^2 \geq 2ab = 2c$, so $c \geq 2$, with equality iff $a = b$, i.e. $a = b = \sqrt{2}$.

For a right triangle the inradius is $r = \dfrac{a + b - c}{2}$. Since $(a + b)^2 = a^2 + b^2 + 2ab = c^2 + 2c$, $r = \dfrac{\sqrt{c^2 + 2c} - c}{2}$.

Rationalising: $r = \dfrac{2c}{2\left(\sqrt{c^2 + 2c} + c\right)} = \dfrac{1}{1 + \sqrt{1 + \frac{2}{c}}}$, which increases as c increases (and tends to $\dfrac{1}{2}$ as $c \to \infty$).

So the least value of r occurs at the least c, namely $c = 2$: $r = \dfrac{1}{1 + \sqrt{2}} = \sqrt{2} - 1 = 0.4142\ldots$

Correct to two decimal places, the minimum value of r is 0.41.

Final Answer: 0.41

Question 22

English

Let x, y, z be positive real numbers such that $x+4y+9z = 18$, and let A and H denote the arithmetic mean and the harmonic mean of x, y, z, respectively. Then the value of $A \cdot H$ at the point (x, y, z) at which H attains its greatest value is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना x, y, z ऐसी धनात्मक वास्तविक संख्याएँ हैं कि $x + 4y + 9z = 18$ है, तथा A और H क्रमशः x, y, z के समान्तर माध्य और हरात्मक माध्य हैं। तब जिस बिन्दु (x, y, z) पर H अपना अधिकतम मान प्राप्त करता है, उस पर $A \cdot H$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$H = \dfrac{3}{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}}$, so H is greatest exactly when $\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}$ is least under $x + 4y + 9z = 18$.

Take the vectors $\left(\sqrt{x}, 2\sqrt{y}, 3\sqrt{z}\right)$ and $\left(\dfrac{1}{\sqrt{x}}, \dfrac{1}{\sqrt{y}}, \dfrac{1}{\sqrt{z}}\right)$. Their dot product is $1 + 2 + 3 = 6$, so by Cauchy-Schwarz $36 \le (x + 4y + 9z)\left(\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}\right) = 18\left(\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}\right)$.

Hence $\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z} \ge 2$ and $H \le \dfrac{3}{2}$.

The same bound follows from AM-GM: expanding, $(x+4y+9z)\left(\dfrac{1}{x}+\dfrac{1}{y}+\dfrac{1}{z}\right) = 14+\left(\dfrac{4y}{x}+\dfrac{x}{y}\right)+\left(\dfrac{9z}{x}+\dfrac{x}{z}\right)+\left(\dfrac{9z}{y}+\dfrac{4y}{z}\right) \ge 14+4+6+12 = 36$.

Equality holds iff the vectors are parallel, i.e. $\sqrt{x}\cdot\sqrt{x} = 2\sqrt{y}\cdot\sqrt{y} = 3\sqrt{z}\cdot\sqrt{z}$, that is, $x = 2y = 3z$ (the same as the AM-GM equality conditions).

With $x = k$, $y = \dfrac{k}{2}$, $z = \dfrac{k}{3}$: $k + 2k + 3k = 18$, so $k = 3$ and the point is $\left(3, \dfrac{3}{2}, 1\right)$, where $H = \dfrac{3}{2}$.

At this point $A = \dfrac{3+\frac{3}{2}+1}{3} = \dfrac{11}{6}$, so $A \cdot H = \dfrac{11}{6}\cdot\dfrac{3}{2} = \dfrac{11}{4} = 2.75$. (Consistently, $A \ge G \ge H$: $G = \sqrt[3]{4.5} \approx 1.65$.)

Final Answer: 2.75

Question 23

English

Let $h_1, h_2, h_3, \ldots$ be an infinite harmonic progression with $h_1 = 60$ and $h_5 = 12$. Then the number of terms of this progression that are integers is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $h_1, h_2, h_3, \ldots$ एक अनन्त हरात्मक श्रेढ़ी है जिसमें $h_1 = 60$ तथा $h_5 = 12$ है। तब इस श्रेढ़ी के उन पदों की संख्या, जो पूर्णांक हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The reciprocals $\frac{1}{h_1}, \frac{1}{h_2}, \ldots$ form an arithmetic progression; let its common difference be D.

$\frac{1}{h_5} = \frac{1}{h_1} + 4D$ gives $\frac{1}{12} = \frac{1}{60} + 4D$, so $4D = \frac{4}{60}$ and $D = \frac{1}{60}$.

Then $\frac{1}{h_n} = \frac{1}{60} + \frac{n-1}{60} = \frac{n}{60}$, so $h_n = \frac{60}{n}$ for every $n \geq 1$ (all terms are defined since $\frac{n}{60} \neq 0$).

$h_n = \frac{60}{n}$ is an integer exactly when n divides 60.

$60 = 2^2 \cdot 3 \cdot 5$ has $(2+1)(1+1)(1+1) = 12$ positive divisors, namely $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$, giving the integer terms $60, 30, 20, 15, 12, 10, 6, 5, 4, 3, 2, 1$.

So exactly 12 terms are integers.

Final Answer: 12

Question 24

English

Let a, b, c be positive real numbers with $a < b < c$ which are in harmonic progression and satisfy $a+b+c = 13$ and $abc = 72$. Let d be the real number for which b, c, d are in harmonic progression. Then the harmonic mean of a and d is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना a, b, c ऐसी धनात्मक वास्तविक संख्याएँ हैं, जहाँ $a < b < c$ है, जो हरात्मक श्रेढ़ी में हैं तथा $a + b + c = 13$ और $abc = 72$ को संतुष्ट करती हैं। माना d वह वास्तविक संख्या है जिसके लिए b, c, d हरात्मक श्रेढ़ी में हैं। तब a और d का हरात्मक माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

In a harmonic progression the middle term is the harmonic mean of its neighbours: $b = \dfrac{2ac}{a+c}$.

From the data $a + c = 13 - b$ and $ac = \dfrac{72}{b}$, so $b = \dfrac{144}{b(13-b)}$, i.e. $b^3 - 13b^2 + 144 = 0$.

$b = 4$ is a root ($64 - 208 + 144 = 0$), and $b^3 - 13b^2 + 144 = (b-4)(b^2 - 9b - 36) = (b-4)(b-12)(b+3)$.

$b = -3$ is rejected since $b > 0$. For $b = 12$: $a + c = 1$ and $ac = 6$, but then a, c would be roots of $t^2 - t + 6 = 0$, which has negative discriminant; rejected.

So $b = 4$, $a + c = 9$, $ac = 18$, and a, c are the roots of $t^2 - 9t + 18 = 0$: with $a < c$, $a = 3$, $c = 6$ (and indeed $3 < 4 < 6$).

The reciprocals $\dfrac{1}{3}, \dfrac{1}{4}, \dfrac{1}{6}$ are in arithmetic progression with common difference $-\dfrac{1}{12}$. For b, c, d in harmonic progression, $\dfrac{1}{d} = \dfrac{2}{c} - \dfrac{1}{b} = \dfrac{1}{3} - \dfrac{1}{4} = \dfrac{1}{12}$, so $d = 12$.

The harmonic mean of $a = 3$ and $d = 12$ is $\dfrac{2 \cdot 3 \cdot 12}{3 + 12} = \dfrac{72}{15} = 4.8$.

Final Answer: 4.8

Question 25

English

Let α, β, γ be the roots of the equation $2x^3 - 6x^2 - 8x + 5 = 0$. Then the harmonic mean of the three numbers $\alpha\beta,\ \beta\gamma,\ \gamma\alpha$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना α, β, γ समीकरण $2x^3 - 6x^2 - 8x + 5 = 0$ के मूल हैं। तब तीन संख्याओं $\alpha\beta,\ \beta\gamma,\ \gamma\alpha$ का हरात्मक माध्य _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The harmonic mean of three non-zero numbers p, q, r is $\dfrac{3}{\frac{1}{p} + \frac{1}{q} + \frac{1}{r}}$.

Here $\dfrac{1}{\alpha\beta} + \dfrac{1}{\beta\gamma} + \dfrac{1}{\gamma\alpha} = \dfrac{\gamma + \alpha + \beta}{\alpha\beta\gamma}$, so the harmonic mean is $\dfrac{3\alpha\beta\gamma}{\alpha + \beta + \gamma}$.

By Vieta's relations for $2x^3 - 6x^2 - 8x + 5 = 0$: $\alpha + \beta + \gamma = \dfrac{6}{2} = 3$ and $\alpha\beta\gamma = -\dfrac{5}{2}$. The product is non-zero, so no root is 0 and every $\alpha\beta$ is non-zero.

(The roots are real and distinct: the cubic changes sign on $(-2, 0)$, $(0, 1)$ and $(1, 5)$.)

Harmonic mean $= \dfrac{3 \cdot \left(-\frac{5}{2}\right)}{3} = -\dfrac{5}{2} = -2.5$.

Final Answer: -2.5

Question 26

English

Let $b_k = 2^{k-4}$ for $k = 1, 2, 3, \ldots$, and for each positive integer n let $P_n = b_1^{1/2}\, b_2^{1/4}\, b_3^{1/8} \cdots b_n^{1/2^n}$. Then the value of $\log_2 \left(\lim\limits_{n\to\infty} P_n \right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $k = 1, 2, 3, \ldots$ के लिए $b_k = 2^{k-4}$ है, तथा प्रत्येक धनात्मक पूर्णांक n के लिए $P_n = b_1^{1/2}\, b_2^{1/4}\, b_3^{1/8} \cdots b_n^{1/2^n}$ है। तब $\log_2 \left(\lim\limits_{n\to\infty} P_n \right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Taking logarithms to base 2: $\log_2 P_n = \displaystyle\sum_{k=1}^{n} \frac{1}{2^k} \log_2 b_k = \sum_{k=1}^{n} \frac{k-4}{2^k}$. So the product becomes the partial sum of the series $\displaystyle\sum \frac{k-4}{2^k}$, whose numerators are in arithmetic progression and denominators in geometric progression.

Split it as $\displaystyle\sum \frac{k}{2^k} - 4 \sum \frac{1}{2^k}$; both series converge.

For $S = \displaystyle\sum_{k\geq 1} \frac{k}{2^k}$: $S - \dfrac{S}{2} = \dfrac{1}{2} + \dfrac{1}{4} + \dfrac{1}{8} + \cdots = 1$, so $S = 2$.

$\sum_{k\geq 1} \frac{4}{2^k} = 4.$

Hence $\log_2 P_n \to 2 - 4 = -2$. Since 2^x is continuous, $P_n \to 2^{-2} = \frac{1}{4}$, and $\log_2(\lim P_n) = -2$.

Final Answer: -2

Question 27

English

The sum $\sum_{n=1}^{\infty} \frac{n^2+1}{n}\left(\frac{3}{4}\right)^n$ can be written as $\alpha + \log_e \beta$, where α and β are positive integers. Then the value of $\alpha\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

योग $\sum_{n=1}^{\infty} \frac{n^2+1}{n}\left(\frac{3}{4}\right)^n$ को $\alpha + \log_e \beta$ के रूप में लिखा जा सकता है, जहाँ α तथा β धनात्मक पूर्णांक हैं। तब $\alpha\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Split the general term: $\frac{n^2+1}{n}x^n = nx^n + \frac{x^n}{n}$ with $x = \frac{3}{4}$; both resulting series converge since $|x| < 1$.

The first is an arithmetico-geometric series: with $S = \sum_{n\geq 1} nx^n$, $S - xS = \sum_{n\geq 1} x^n = \frac{x}{1-x}$, so $S = \frac{x}{(1-x)^2} = \frac{3/4}{1/16} = 12.$

The second is the logarithmic series: $\log_e(1-x) = -\sum_{n\geq 1} \frac{x^n}{n}$ for $|x| < 1$, so $\sum_{n\geq 1} \frac{(3/4)^n}{n} = -\log_e \frac{1}{4} = \log_e 4$.

Hence the sum equals $12 + \log_e 4$.

Uniqueness: if $\alpha + \log_e \beta = 12 + \log_e 4$ with positive integers α, β, then $\beta = 4e^{12-\alpha}$. For an integer $k \neq 0$, e^k is irrational, so β is an integer only when $\alpha = 12$, and then $\beta = 4$.

$\alpha\beta = 12 \cdot 4 = 48.$

Final Answer: 48

Question 28

English

The value of the series $\sum_{n=1}^{\infty} \frac{n(n+1)}{3^{n+1}} = \frac{2}{3^2} + \frac{6}{3^3} + \frac{12}{3^4} + \frac{20}{3^5} + \cdots$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

श्रेणी $\sum_{n=1}^{\infty} \frac{n(n+1)}{3^{n+1}} = \frac{2}{3^2} + \frac{6}{3^3} + \frac{12}{3^4} + \frac{20}{3^5} + \cdots$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $S = \sum_{n\geq 1} \frac{n(n+1)}{3^{n+1}}$; it converges (ratio test with ratio $\frac{1}{3}$).

$$\frac{S}{3} = \sum_{n\geq 1} \frac{n(n+1)}{3^{n+2}} = \sum_{n\geq 2} \frac{(n-1)n}{3^{n+1}}.$$

Subtracting term by term: $S - \frac{S}{3} = \sum_{n\geq 1} \frac{n(n+1)-(n-1)n}{3^{n+1}} = \sum_{n\geq 1} \frac{2n}{3^{n+1}}$, since the numerators $2, 6, 12, 20, \ldots$ have differences $4, 6, 8, \cdots = 2(n)$, an arithmetic progression. The new series is arithmetico-geometric.

$\sum_{n\geq 1} \frac{2n}{3^{n+1}} = \frac{2}{3} \sum_{n\geq 1} \frac{n}{3^n}$, and with $T = \sum \frac{n}{3^n}$: $T - \frac{T}{3} = \sum \frac{1}{3^n} = \frac{1}{2}$, so $T = \frac{3}{4}$.

Hence $\frac{2}{3}S = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$ and $S = \frac{3}{4} = 0.75$.

Final Answer: 0.75

Question 29

English

Let $u_1, u_2, u_3, \ldots$ be the sequence defined by $u_1 = 2$, $u_2 = 5$ and $u_{n+2} - 2u_{n+1} + u_n = 2 + 2^n$ for all $n \geq 1$. Then the value of $u_1 + u_2 + \cdots + u_8$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $u_1, u_2, u_3, \ldots$, $u_1 = 2$, $u_2 = 5$ तथा सभी $n \geq 1$ के लिए $u_{n+2} - 2u_{n+1} + u_n = 2 + 2^n$ द्वारा परिभाषित अनुक्रम है। तब $u_1 + u_2 + \cdots + u_8$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $v_n = u_{n+1} - u_n$. The given relation says $v_{n+1} - v_n = 2 + 2^n$, and $v_1 = u_2 - u_1 = 3$.

Adding: $v_n = v_1 + \sum_{k=1}^{n-1}(2 + 2^k) = 3 + 2(n-1) + (2^n - 2) = 2^n + (2n - 1)$.

So the differences of consecutive terms split into $2n - 1$ (an arithmetic progression) and 2^n (a geometric progression).

Then $u_n = u_1 + \sum_{k=1}^{n-1} v_k = 2 + \sum_{k=1}^{n-1} 2^k + \sum_{k=1}^{n-1}(2k - 1) = 2 + (2^n - 2) + (n-1)^2 = 2^n + (n-1)^2$.

Check: $u_1 = 2$, $u_2 = 5$, $u_3 = 12$, $u_4 = 25$, $u_5 = 48$, and $u_3 - 2u_2 + u_1 = 4 = 2 + 2^1$.

$$\sum_{n=1}^{8} u_n = \sum_{n=1}^{8} 2^n + \sum_{j=0}^{7} j^2 = (2^9 - 2) + \frac{7 \cdot 8 \cdot 15}{6} = 510 + 140 = 650.$$

Final Answer: 650

Question 30

English

Consider the sequence $4, 5, 8, 17, 44, 125, \ldots$ in which the differences of consecutive terms, $1, 3, 9, 27, 81, \ldots$, form a geometric progression. Then the least positive integer n for which the n-th term of this sequence exceeds 10^6 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

अनुक्रम $4, 5, 8, 17, 44, 125, \ldots$ पर विचार कीजिए, जिसमें क्रमागत पदों के अन्तर $1, 3, 9, 27, 81, \ldots$ एक गुणोत्तर श्रेढ़ी बनाते हैं। तब वह न्यूनतम धनात्मक पूर्णांक n, जिसके लिए इस अनुक्रम का n-वाँ पद 10^6 से अधिक है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let u_n be the n-th term. Then $u_n = u_1 + \sum_{k=1}^{n-1}(u_{k+1} - u_k) = 4 + \sum_{k=1}^{n-1} 3^{k-1}$.

The sum is a finite GP: $\sum_{k=1}^{n-1} 3^{k-1} = \dfrac{3^{n-1}-1}{2}$, so $u_n = 4 + \dfrac{3^{n-1}-1}{2}$ (check: $u_6 = 4 + 121 = 125$).

u_n is increasing, and $u_n > 10^6 \iff \dfrac{3^{n-1}-1}{2} > 999996 \iff 3^{n-1} > 1999993$.

Powers of 3: $3^{13} = 1594323$ and $3^{14} = 4782969$. So $3^{n-1} > 1999993$ first holds for $n - 1 = 14$.

Hence the least n is 15; indeed $u_{14} = 4 + 797161 = 797165 < 10^6$ and $u_{15} = 4 + 2391484 = 2391488 > 10^6$.

Final Answer: 15

Question 31

English

For each positive integer n, let $T_n = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1)$. Then the value of $\sum_{n=1}^{\infty} \frac{n+3}{T_n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना $T_n = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1)$ है। तब $\sum_{n=1}^{\infty} \frac{n+3}{T_n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each term of T_n telescopes: $k(k+1) = \frac{k(k+1)(k+2) - (k-1)k(k+1)}{3}$, so $T_n = \frac{n(n+1)(n+2)}{3}$.

Hence $\frac{n+3}{T_n} = \frac{3(n+3)}{n(n+1)(n+2)}$. Writing $n+3 = (n+2)+1$: $\frac{n+3}{n(n+1)(n+2)} = \frac{1}{n(n+1)} + \frac{1}{n(n+1)(n+2)}$.

$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so $\sum_{n=1}^{N} \frac{1}{n(n+1)} = 1 - \frac{1}{N+1} \to 1$.

$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$, so $\sum_{n=1}^{N} \frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right] \to \frac{1}{4}$.

Therefore $\sum_{n=1}^{\infty} \frac{n+3}{T_n} = 3\left(1 + \frac{1}{4}\right) = \frac{15}{4} = 3.75$.

Final Answer: 3.75

Question 32

English

For each positive integer n, let $x_n = \sum_{j=1}^{n} \frac{2}{(2j-1)(2j+1)}$, and let $S = \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{x_{n+1}-x_n}{1+x_n x_{n+1}}\right)$, where $\tan^{-1}$ takes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the value of $\tan S$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना $x_n = \sum_{j=1}^{n} \frac{2}{(2j-1)(2j+1)}$ है, तथा $S = \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{x_{n+1}-x_n}{1+x_n x_{n+1}}\right)$ है, जहाँ $\tan^{-1}$ के मान $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में हैं। तब $\tan S$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Partial fractions: $\frac{2}{(2j-1)(2j+1)} = \frac{1}{2j-1} - \frac{1}{2j+1}$, so the sum telescopes and $x_n = 1 - \frac{1}{2n+1} = \frac{2n}{2n+1}$.

Thus $x_1 = \frac{2}{3}$, the x_n increase, $0 < x_n < 1$, and $x_n \to 1$.

For a, b with $ab > -1$, $\tan^{-1} a - \tan^{-1} b = \tan^{-1} \frac{a-b}{1+ab}$ (principal values). Here $x_n x_{n+1} > 0$, so each term equals $\tan^{-1} x_{n+1} - \tan^{-1} x_n$, a positive number less than $\frac{\pi}{4}$.

The partial sums telescope: $\sum_{n=1}^{N} \left(\tan^{-1} x_{n+1} - \tan^{-1} x_n\right) = \tan^{-1} x_{N+1} - \tan^{-1} x_1 \to \tan^{-1} 1 - \tan^{-1} \frac{2}{3}$.

So $S = \frac{\pi}{4} - \tan^{-1} \frac{2}{3}$ and $\tan S = \frac{1 - \frac{2}{3}}{1 + 1 \cdot \frac{2}{3}} = \frac{1/3}{5/3} = \frac{1}{5} = 0.2$.

Final Answer: 0.2

Question 33

English

The sum $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\,4}{n(n+1)(n+2)}$ can be written as $\alpha \log_e 2 + \beta$, where α and β are integers. Then the value of $\alpha\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

योग $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\,4}{n(n+1)(n+2)}$ को $\alpha \log_e 2 + \beta$ के रूप में लिखा जा सकता है, जहाँ α तथा β पूर्णांक हैं। तब $\alpha\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Partial fractions: $\frac{4}{n(n+1)(n+2)} = \frac{2}{n} - \frac{4}{n+1} + \frac{2}{n+2}$ (the weights $1, -2, 1$ that make the non-alternating series telescope).

The series $\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ at $x = 1$ gives $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} = \log_e 2$.

Shifting the index: $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n+1} = \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \cdots = 1 - \log_e 2$, and $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n+2} = \frac{1}{3} - \frac{1}{4} + \cdots = \log_e 2 - 1 + \frac{1}{2} = \log_e 2 - \frac{1}{2}$.

All three series converge, so the given sum is $2\log_e 2 - 4(1 - \log_e 2) + 2\left(\log_e 2 - \frac{1}{2}\right) = 8\log_e 2 - 5$.

Numerically $8\log_e 2 - 5 \approx 0.5452$, which agrees with the partial sums.

If $\alpha \log_e 2 + \beta = 8\log_e 2 - 5$ with integers α, β and $\alpha \neq 8$, then $\log_e 2$ would be rational, which is false. So $\alpha = 8$, $\beta = -5$, and $\alpha\beta = -40$.

Final Answer: -40

Question 34

English

For each positive integer n, let $A_n = \sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)}$, $B_n = \sum_{k=1}^{n} k(k+1)(k+2)$ and $C_n = \sum_{k=1}^{n} \frac{k}{k^4+k^2+1}$. Let N be the positive integer for which $A_N = \frac{5}{21}$. Then the value of $\frac{B_N}{C_N}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना $A_n = \sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)}$, $B_n = \sum_{k=1}^{n} k(k+1)(k+2)$ तथा $C_n = \sum_{k=1}^{n} \frac{k}{k^4+k^2+1}$ है। माना N वह धनात्मक पूर्णांक है जिसके लिए $A_N = \frac{5}{21}$ है। तब $\frac{B_N}{C_N}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Telescoping with linear factors: $\frac{1}{k(k+1)(k+2)} = \frac{1}{2}\left[\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}\right]$, so $A_n = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right] = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}$.

$A_N = \frac{5}{21}$ gives $\frac{1}{2(N+1)(N+2)} = \frac{1}{4} - \frac{5}{21} = \frac{1}{84}$, so $(N+1)(N+2) = 42 = 6 \cdot 7$ and $N = 5$ (unique, since A_n is increasing).

Product of linear factors: $k(k+1)(k+2) = \frac{k(k+1)(k+2)(k+3) - (k-1)k(k+1)(k+2)}{4}$, so $B_n = \frac{n(n+1)(n+2)(n+3)}{4}$ and $B_5 = \frac{5 \cdot 6 \cdot 7 \cdot 8}{4} = 420$.

For C_n: $k^4 + k^2 + 1 = (k^2+1)^2 - k^2 = (k^2 - k + 1)(k^2 + k + 1)$, and the two factors differ by $2k$, so $\frac{k}{k^4+k^2+1} = \frac{1}{2}\left[\frac{1}{k^2-k+1} - \frac{1}{k^2+k+1}\right]$.

Since $k^2 + k + 1 = (k+1)^2 - (k+1) + 1$, the bracket is $f(k) - f(k+1)$ with $f(k) = \frac{1}{k^2-k+1}$, so $C_n = \frac{1}{2}\left(f(1) - f(n+1)\right) = \frac{1}{2}\left(1 - \frac{1}{n^2+n+1}\right)$.

$C_5 = \frac{1}{2}\left(1 - \frac{1}{31}\right) = \frac{15}{31}$.

$\frac{B_5}{C_5} = 420 \cdot \frac{31}{15} = 28 \cdot 31 = 868$.

Final Answer: 868

Question 35

English

Let $\theta = \tan^{-1}\frac{1}{3}$, where $\tan^{-1}$ takes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. For each positive integer n, let $E_n = \sum_{k=1}^{n} \tan(k\theta)\tan\left((k+1)\theta\right)$ and $F_n = \sum_{k=1}^{n} k(k+1)$. Then the value of $\frac{E_4}{F_4}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $\theta = \tan^{-1}\frac{1}{3}$ है, जहाँ $\tan^{-1}$ के मान $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में हैं। प्रत्येक धनात्मक पूर्णांक n के लिए माना $E_n = \sum_{k=1}^{n} \tan(k\theta)\tan\left((k+1)\theta\right)$ तथा $F_n = \sum_{k=1}^{n} k(k+1)$ है। तब $\frac{E_4}{F_4}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $t_k = \tan k\theta$, so $t_1 = \frac{1}{3}$. Since $\theta = (k+1)\theta - k\theta$, $\tan\theta = \frac{t_{k+1} - t_k}{1 + t_k t_{k+1}}$, i.e. $t_k t_{k+1} = \frac{t_{k+1} - t_k}{\tan\theta} - 1 = 3(t_{k+1} - t_k) - 1$ (valid for $k \le 4$, since $5\theta \neq \frac{\pi}{2}$).

Summing, the differences telescope: $E_n = 3(t_{n+1} - t_1) - n$.

$t_2 = \frac{2t_1}{1 - t_1^2} = \frac{2/3}{8/9} = \frac{3}{4}$; $t_4 = \frac{2t_2}{1 - t_2^2} = \frac{3/2}{7/16} = \frac{24}{7}$; $t_5 = \frac{t_4 + t_1}{1 - t_4 t_1} = \frac{\frac{24}{7} + \frac{1}{3}}{1 - \frac{8}{7}} = \frac{79/21}{-1/7} = -\frac{79}{3}$.

$E_4 = 3\left(-\frac{79}{3} - \frac{1}{3}\right) - 4 = -80 - 4 = -84$. (Direct check with $t_3 = \frac{13}{9}$: $\frac{1}{4} + \frac{13}{12} + \frac{104}{21} - \frac{632}{7} = -84$.)

For the product of linear factors: $k(k+1) = \frac{k(k+1)(k+2) - (k-1)k(k+1)}{3}$ telescopes to $F_n = \frac{n(n+1)(n+2)}{3}$, so $F_4 = \frac{4 \cdot 5 \cdot 6}{3} = 40$.

$\frac{E_4}{F_4} = -\frac{84}{40} = -2.1$.

Final Answer: -2.1

Question 36

English

Let $n = 9999$, and let $A = \sum_{k=1}^{n} \frac{1}{\sqrt{k} + \sqrt{k+1}}$ and $B = \sum_{k=1}^{n} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$. Then the value of $\log_{10}\left(\frac{B}{A}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $n = 9999$ है, तथा $A = \sum_{k=1}^{n} \frac{1}{\sqrt{k} + \sqrt{k+1}}$ और $B = \sum_{k=1}^{n} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$ है। तब $\log_{10}\left(\frac{B}{A}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Multiplying by the conjugate, $\frac{1}{\sqrt{k} + \sqrt{k+1}} = \frac{\sqrt{k+1} - \sqrt{k}}{(k+1) - k} = \sqrt{k+1} - \sqrt{k}$, so A telescopes: $A = \sqrt{n+1} - 1$.

For B, factor the denominator: $(k+1)\sqrt{k} + k\sqrt{k+1} = \sqrt{k}\sqrt{k+1}\left(\sqrt{k+1} + \sqrt{k}\right)$.

Hence the k-th term of B is $\frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k}\sqrt{k+1}} = \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}$, and $B = 1 - \frac{1}{\sqrt{n+1}} = \frac{\sqrt{n+1} - 1}{\sqrt{n+1}}$.

So $\frac{B}{A} = \frac{1}{\sqrt{n+1}}$ for every n.

With $n = 9999$: $\sqrt{n+1} = 100$, $A = 99$, $B = 0.99$, and $\log_{10} \frac{B}{A} = \log_{10} 10^{-2} = -2$.

Final Answer: -2

Question 37

English

Let m be the positive integer such that $\sum_{k=1}^{5} \frac{k^2+k-1}{(k+2)!} = \frac{1}{2} - \frac{1}{m}$. Then the value of m is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना m वह धनात्मक पूर्णांक है जिसके लिए $\sum_{k=1}^{5} \frac{k^2+k-1}{(k+2)!} = \frac{1}{2} - \frac{1}{m}$ है। तब m का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Look for $g(k) = \frac{k}{(k+1)!}$ with $g(k) - g(k+1)$ equal to the term: $\frac{k}{(k+1)!} - \frac{k+1}{(k+2)!} = \frac{k(k+2)-(k+1)}{(k+2)!} = \frac{k^2+k-1}{(k+2)!}$.

So $\sum_{k=1}^{n} \frac{k^2+k-1}{(k+2)!} = g(1) - g(n+1) = \frac{1}{2!} - \frac{n+1}{(n+2)!}$.

For $n = 5$: $\frac{6}{7!} = \frac{6}{5040} = \frac{1}{840}$, so the sum is $\frac{1}{2} - \frac{1}{840}$.

Check by direct addition: $\frac{1}{6} + \frac{5}{24} + \frac{11}{120} + \frac{19}{720} + \frac{29}{5040} = \frac{419}{840} = \frac{1}{2} - \frac{1}{840}$.

Hence $m = 840$.

Final Answer: 840

Question 38

English

Let $u_1, u_2, u_3, \ldots$ be the sequence defined by $u_1 = 1$ and $u_{n+1} = 4u_n + 3^n$ for all $n \geq 1$. Then the value of $\sum_{n=1}^{\infty} \frac{u_n}{8^n}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना $u_1, u_2, u_3, \ldots$, $u_1 = 1$ तथा सभी $n \geq 1$ के लिए $u_{n+1} = 4u_n + 3^n$ द्वारा परिभाषित अनुक्रम है। तब $\sum_{n=1}^{\infty} \frac{u_n}{8^n}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The forcing term 3^n suggests adding a multiple of 3^n to u_n. With $v_n = u_n + 3^n$: $v_{n+1} = u_{n+1} + 3^{n+1} = 4u_n + 3^n + 3 \cdot 3^n = 4(u_n + 3^n) = 4v_n$.

So (v_n) is a geometric progression with ratio 4 and $v_1 = 1 + 3 = 4$; hence $v_n = 4^n$ and $u_n = 4^n - 3^n$.

Check: $u_1 = 1$, $u_2 = 4 + 3 = 7 = 16 - 9$, $u_3 = 28 + 9 = 37 = 64 - 27$.

$\sum_{n\geq 1} \frac{u_n}{8^n} = \sum_{n\geq 1} \left(\frac{1}{2}\right)^n - \sum_{n\geq 1} \left(\frac{3}{8}\right)^n$; both geometric series converge.

$\sum \left(\frac{1}{2}\right)^n = 1$ and $\sum \left(\frac{3}{8}\right)^n = \frac{3/8}{1 - 3/8} = \frac{3}{5}$.

The required value is $1 - \frac{3}{5} = \frac{2}{5} = 0.4$.

Final Answer: 0.4

Question 39

English

Two players A and B share a ball, which is with A at time $t = 0$. At each of the times $t = 1, 2, 3, \ldots$, whoever holds the ball keeps it with probability $\frac{4}{5}$ and passes it to the other player with probability $\frac{1}{5}$, independently of all earlier times. For each positive integer n, let p_n and q_n denote the probabilities that the ball is with A and with B, respectively, just after time $t = n$. Then the value of $\sum_{n=1}^{\infty} (p_n - q_n)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दो खिलाड़ी A तथा B एक गेंद साझा करते हैं, जो समय $t = 0$ पर A के पास है। प्रत्येक समय $t = 1, 2, 3, \ldots$ पर, जिसके पास गेंद है वह उसे प्रायिकता $\frac{4}{5}$ से अपने पास रखता है तथा प्रायिकता $\frac{1}{5}$ से दूसरे खिलाड़ी को दे देता है, सभी पूर्ववर्ती समयों से स्वतन्त्र रूप से। प्रत्येक धनात्मक पूर्णांक n के लिए माना p_n तथा q_n क्रमशः वे प्रायिकताएँ हैं कि समय $t = n$ के ठीक बाद गेंद A के पास तथा B के पास है। तब $\sum_{n=1}^{\infty} (p_n - q_n)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The ball is with A after time $n+1$ if it was with A and was kept (probability $\frac{4}{5}$), or was with B and was passed (probability $\frac{1}{5}$). So $p_{n+1} = \frac{4}{5}p_n + \frac{1}{5}q_n$ with $q_n = 1 - p_n$, i.e. $p_{n+1} = \frac{1}{5} + \frac{3}{5}p_n$, and $p_0 = 1$.

The fixed point of $x \mapsto \frac{1}{5} + \frac{3}{5}x$ is $x = \frac{1}{2}$. Subtracting it: $p_{n+1} - \frac{1}{2} = \frac{3}{5}\left(p_n - \frac{1}{2}\right)$, so $p_n - \frac{1}{2}$ is a geometric progression with first value $p_0 - \frac{1}{2} = \frac{1}{2}$ and ratio $\frac{3}{5}$.

Hence $p_n = \frac{1}{2} + \frac{1}{2}\left(\frac{3}{5}\right)^n$ and $q_n = \frac{1}{2} - \frac{1}{2}\left(\frac{3}{5}\right)^n$.

$p_n - q_n = \left(\frac{3}{5}\right)^n$. (Check: $p_1 - q_1 = \frac{4}{5} - \frac{1}{5} = \frac{3}{5}$.)

$$\sum_{n=1}^{\infty} (p_n - q_n) = \sum_{n\geq 1} \left(\frac{3}{5}\right)^n = \frac{3/5}{1 - 3/5} = \frac{3}{2} = 1.5.$$

Final Answer: 1.5

Question 40

English

A line $y = mx + c$ meets the curve $y = x^3 - 6x^2 + 5x + 3$ at three distinct points whose x-coordinates are in arithmetic progression, and the sum of the squares of these x-coordinates is 20. Then the value of $m^2 + c^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक रेखा $y = mx + c$ वक्र $y = x^3 - 6x^2 + 5x + 3$ को तीन भिन्न बिन्दुओं पर मिलती है जिनके x-निर्देशांक समान्तर श्रेढ़ी में हैं, तथा इन x-निर्देशांकों के वर्गों का योग 20 है। तब $m^2 + c^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At a common point $x^3 - 6x^2 + 5x + 3 = mx + c$, so the three x-coordinates are the roots of $x^3 - 6x^2 + (5 - m)x + (3 - c) = 0$, and by Vieta their sum is 6.

Write the roots, which are in arithmetic progression, as $a - d$, a, $a + d$. The sum of these three terms is $3a$ (three times the middle term), so $3a = 6$ and $a = 2$.

Sum of squares: $(2 - d)^2 + 4 + (2 + d)^2 = 12 + 2d^2 = 20$, so $d^2 = 4$ and the roots are $0, 2, 4$ (distinct, as required).

Hence $x^3 - 6x^2 + (5 - m)x + (3 - c) = x(x - 2)(x - 4) = x^3 - 6x^2 + 8x$. Comparing coefficients, $5 - m = 8$ and $3 - c = 0$, so $m = -3$ and $c = 3$.

Check: the line $y = -3x + 3$ meets the curve at $(0, 3)$, $(2, -3)$, $(4, -9)$; the middle point $(2, -3)$ is the point of inflection of the cubic.

$m^2 + c^2 = 9 + 9 = 18$.

Final Answer: 18